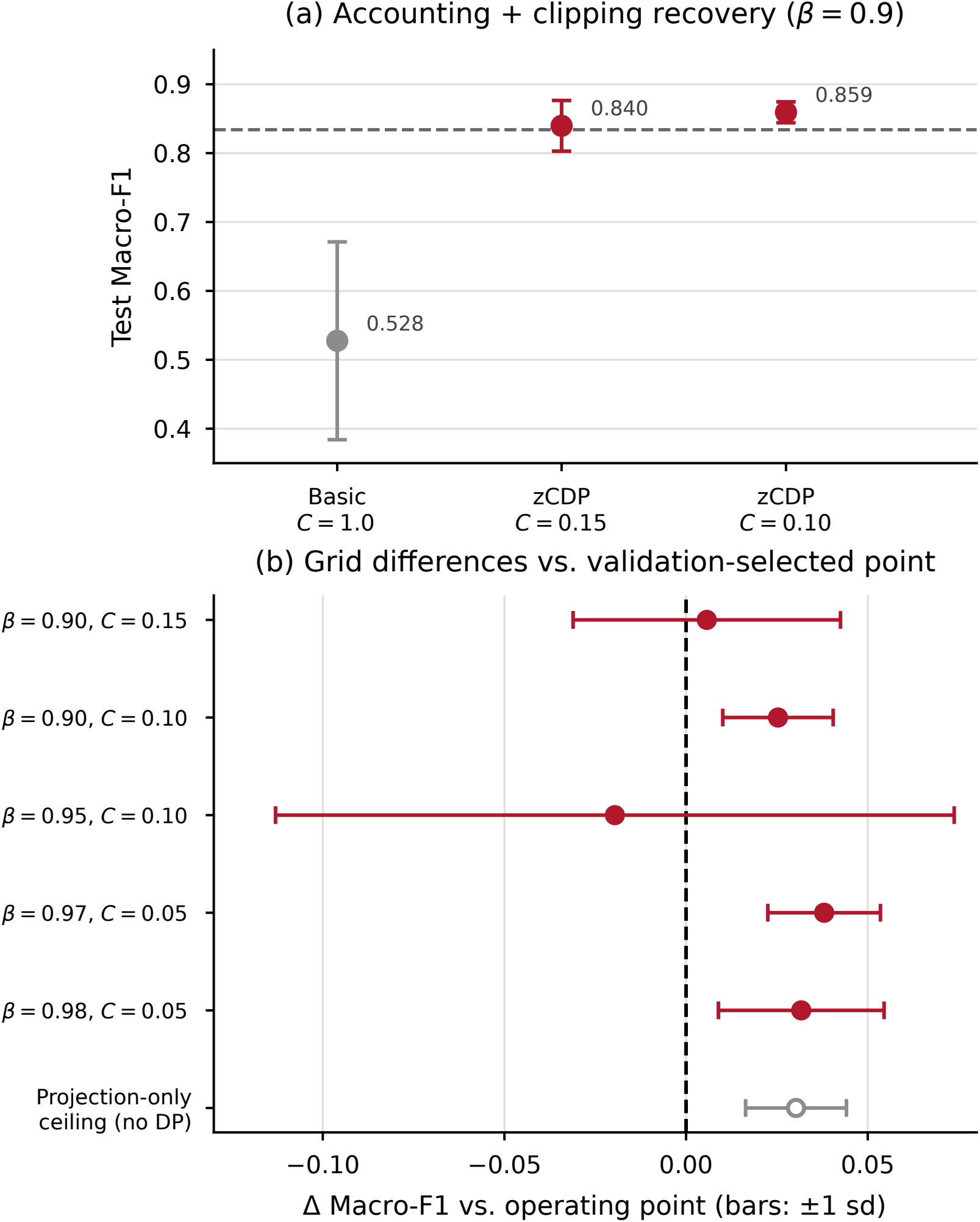
(a) Accounting + clipping recovery ($\beta = 0.9$)
Test Macro-F1
0.9
0.8
0.7
0.6
0.5
0.4
0.840
0.859
0.528
Basic
$C = 1.0$
zCDP
$C = 0.15$
zCDP
$C = 0.10$
(b) Grid differences vs. validation-selected point
$\beta = 0.90, C = 0.15$
$\beta = 0.90, C = 0.10$
$\beta = 0.95, C = 0.10$
$\beta = 0.97, C = 0.05$
$\beta = 0.98, C = 0.05$
Projection-only ceiling (no DP)
−0.10
−0.05
0.00
0.05
Δ Macro-F1 vs. operating point (bars: ±1 sd)